

PKU–Zurich PhD Summer School on Machine Learning for Macroeconomics and Finance

Surrogate-based simulated method of moments in the Brock–Mirman model

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July 6–10, 2026 · Beijing, China

https://liboecon.com/2026_summer_school.html

Roadmap of This Lecture

1. **The model.** Brock–Mirman with stochastic TFP, calibration, what is observed.
2. **SMM primer.** Moment condition, weighting matrix, asymptotic distribution.
3. **Why surrogates.** Direct simulation of the SMM objective is too expensive; replace the inner solver with a trained network.
4. **Task 1: Train surrogate** with ϱ as pseudo-state, Euler-residual loss, Gauss–Hermite quadrature.
5. **Task 2: Compute identifying moments** from surrogate-driven simulations.
6. **Task 3: Estimate $\hat{\varrho}$** via SMM under common random numbers.
7. **Task 4: Joint (β, ϱ) estimation** – identification ridge, over- vs just-identified moment sets.

Learning objectives

By the end of this lecture you should be able to:

- ▶ Build and train a neural surrogate with pseudo-state parameters to replace expensive inner-loop solvers
- ▶ Compute identifying moments from surrogate-driven simulations under common random numbers
- ▶ Implement the SMM estimator using differentiable surrogates and verify interior solutions
- ▶ Diagnose parameter identification through SMM-criterion profiles and moment sensitivity
- ▶ Extend single-parameter SMM to joint estimation and visualize the identification ridge

Where Session 1 Left Us

The surrogate idea

Train a network $\mathcal{N}(x, \theta)$ *once* on the joint state–parameter space (parameters as **pseudo-states**); afterwards, every counterfactual, moment, or derivative is a cheap forward pass.

Session 1 built the surrogate and previewed its three uses:

1. **Estimate** — point an optimizer at the surrogate's moment predictions. (**today**)
2. *Quantify* — propagate parameter uncertainty through the surrogate. (Session 3, appendix)
3. *Design* — search the policy-parameter space for an optimum. (Session 3)

This session works out the **estimate** operator in full: simulated method of moments on a Brock–Mirman surrogate — first a sharply identified scalar (ϱ), then a partially identified pair (β, ϱ) . The Gaussian processes of Lecture 6 return in Session 3 as the *second* surrogate.

Notation. On Day 2, θ denoted network weights; from here on θ is the *structural* parameter vector — the network's weights stay implicit.

Hands-on: 02_01_Structural_Estimation_BM.ipynb (scalar) and 02_02_Structural_Estimation_BM_Joint.ipynb (joint).

SMM Primer (1/3): The Moment Condition

Let the data generate T observations $\{y_t\}$. Pick a vector of **moment functions** $g(y) \in \mathbb{R}^q$ such that, under the true parameter $\theta_0 \in \mathbb{R}^p$ ($q \geq p$),

$$\mathbb{E}[g(y_t) - m(\theta_0)] = 0,$$

where $m(\theta) = \mathbb{E}_\theta[g(y_t)]$ are the **model-implied moments**. The empirical and simulated counterparts are

$$\hat{g}_T = \frac{1}{T} \sum_{t=1}^T g(y_t), \quad \hat{m}_S(\theta) = \frac{1}{S} \sum_{s=1}^S g(y_s^{\text{sim}}(\theta)).$$

Identification. θ_0 is the unique zero of $\mathbb{E}[g(y_t)] - m(\theta)$ on the parameter space. Just-identified: $q = p$. Over-identified: $q > p$ (testable restrictions).

This lecture. y_t = log output / consumption / capital. g = three autocorrelation and volatility moments. $\theta = \varrho$ (Tasks 1–3) or $\theta = (\beta, \varrho)$ (Task 4).

SMM Primer (2/3): Estimator and Weighting

The SMM estimator solves

$$\hat{\theta}_{\text{SMM}} = \arg \min_{\theta} [\hat{g}_T - \hat{m}_S(\theta)]^\top W [\hat{g}_T - \hat{m}_S(\theta)],$$

where W is a positive-definite **weighting matrix**.

Common random numbers (CRN). Hold the random draws $\{\varepsilon_s\}$ fixed across θ . Then $\hat{m}_S(\theta)$ is a smooth function of θ – simulation noise no longer disrupts the optimizer.

Weighting matrix choice.

- ▶ **Identity** $W = I$: easy, but inefficient.
- ▶ **Diagonal** $W = \text{diag}(1/\hat{\sigma}_g^2)$: rescales each moment by its sampling variance.
- ▶ **Optimal** $W^* = \hat{\Sigma}_g^{-1}$ where $\hat{\Sigma}_g$ is the long-run covariance of $g(y_t)$. Achieves minimum asymptotic variance.

SMM Primer (3/3): Asymptotic Distribution

Notation follows the script: $M = \partial m(\theta_0)/\partial \theta^\top$ is the moment Jacobian, Σ_m the long-run covariance of the moment vector, and $\Omega = (1 + 1/(\tau S))\Sigma_m$ the asymptotic covariance of $\sqrt{T}\hat{g}(\theta_0)$ under S independent simulated panels of length τT relative to the data ($\tau \geq 1$). With S panels of the same length as the data ($\tau = 1$), $\Omega = (1 + 1/S)\Sigma_m$; as $\tau S \rightarrow \infty$, $\Omega \rightarrow \Sigma_m$.

Under standard regularity conditions (smoothness of m , stationarity, identification), as the data length $T \rightarrow \infty$,

$$\sqrt{T}(\hat{\theta}_{\text{SMM}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, V(W)),$$
$$V(W) = (M^\top W M)^{-1} M^\top W \Omega W M (M^\top W M)^{-1}.$$

Optimal weighting. With $W = W^* = \Omega^{-1}$,

$$V(W^*) = (M^\top \Omega^{-1} M)^{-1}.$$

Identification, geometrically. M has full column rank $\Rightarrow$ the moment surface bends sharply along every parameter direction. A flat direction = an *identification ridge* (Task 4 visualizes one).

Why the surrogate? M and m require simulating the model. With a deep surrogate the inner loop is differentiable and three orders of magnitude cheaper.

The Model: Brock–Mirman with Stochastic TFP

Stochastic growth model:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \beta^t \mathbb{E}[\ln C_t], \quad K_{t+1} + C_t = Y_t + (1 - \delta)K_t$$
$$Y_t = z_t K_t^\alpha, \quad \log z_{t+1} = \varrho \log z_t + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1).$$

Calibrated parameters

α	δ	σ	β (single- ϱ exercise)
0.36	0.10	0.04	0.96

Note: the textbook closed-form policy $s = \alpha\beta$ requires $\delta = 1$. We use $\delta = 0.10$, so no analytic policy exists and the Euler equation must be solved numerically – the surrogate replaces this expensive inner solve.

Pedagogical sequence:

- ▶ **Exercise 1:** estimate $\varrho \in [0.50, 0.99]$ from autocorrelation moments. Sharp 1D identification.
- ▶ **Exercise 2:** estimate (β, ϱ) *jointly*. Visualize identification ridge; collapse it with extra moments.

Why ϱ , not β ?

- ▶ β is famously **poorly identified** from macro moments alone – it shifts the level of the savings rate but every level moment depends on the same scalar.
- ▶ In practice, β is calibrated to a target steady-state interest rate (under log utility and full depreciation, $1/\beta = 1 + r$ at SS; with general δ , the SS Euler relation is $1/\beta = 1 + \alpha(k^*)^{\alpha-1} - \delta$).
- ▶ ϱ has a **clean direct mapping** to autocorrelation moments: $\text{Corr}(\log Y_t, \log Y_{t-1})$, $\text{Corr}(\Delta \log C_t, \Delta \log C_{t-1})$, and the simulated volatility moment move with persistence.
- ▶ This is the textbook example of a **sharply identified** scalar parameter – exactly what we want for teaching the SMM workflow. (With three moments and one parameter the system is technically *over-identified*; the extra moments make the criterion sharper.)

The single-parameter exercise (notebook 02_01) recovers $\hat{\varrho}$ accurately under CRN. The joint exercise (notebook 02_02) shows what happens when an additional, partially-identified parameter is freed.

Task 1: Train Surrogate with ϱ as Pseudo-State

Goal

Train $\mathcal{N}(z_t, K_t, \varrho) \rightarrow s_t \in (0, 1)$, with $K_{t+1} = (1 - \delta)K_t + Y_t \cdot s_t$.

1. 3-input MLP $(z, K, \varrho) \rightarrow s$ with sigmoid output.
2. Euler residual loss with Gauss–Hermite quadrature; ϱ enters the *transition* as well as the policy. Under log utility, $u'(C) = 1/C$; the notebooks minimize the *relative* (scale-free) residual, which shares the zero set of $1 - \beta C_t \mathbb{E}[\cdot]$ but is better conditioned in FP32:

$$\ell = \frac{1}{N} \sum_i \left(\left[\beta C_t^{(i)} \sum_j w_j \frac{1 - \delta + \alpha z_{t+1,j}^{(i)} K_{t+1}^{(i)\alpha-1}}{C_{t+1,j}^{(i)}} \right]^{-1} - 1 \right)^2$$

3. Two-phase training: (i) uniform sampling on (z, K, ϱ) box, (ii) simulation-based refinement.
4. Validate by plotting $s(z = 1, K, \varrho)$ for several ϱ .

Task 2: Simulate and Compute Identifying Moments

Goal

Generate synthetic “ex-post” data at $\varrho_{\text{true}} = 0.90$ and compute three identifying moments (plus one diagnostic, see below).

Moment	What it captures
$m_1 = \text{Std}(\Delta \log C_t)$	Growth-rate volatility; check numerically
$m_2 = \text{Corr}(\Delta \log C_t, \Delta \log C_{t-1})$	Persistence in consumption growth
$m_3 = \text{Corr}(\log Y_t, \log Y_{t-1})$	Persistence in output (very direct)
$m_0 = \text{mean}(s_t)$ (<i>diagnostic only</i>)	Should be flat in ϱ ; included to show what <i>not</i> to use

Note: $\text{Var}(\log z) = \sigma^2/(1 - \varrho^2)$, but $\text{Var}(\Delta \log z) = 2\sigma^2/(1 + \varrho)$.

CRN: same burn-in, horizon, initial state, seed for every candidate ϱ .

Task 3: Estimate ϱ via SMM

SMM objective (single-parameter case, $\theta = \varrho$)

$$\hat{\varrho} = \arg \min_{\varrho} [m(\varrho) - \hat{m}^{\text{data}}]^{\top} W [m(\varrho) - \hat{m}^{\text{data}}], \quad W = I.$$

$W = I$ is the unweighted criterion – the cleanest pedagogical default. The efficient choice is $W^* = \hat{\Sigma}_g^{-1}$, the inverse long-run covariance of the moment vector (the $\hat{\Sigma}_g$ of the SMM-primer slides); with $W = I$ the asymptotic variance of $\hat{\varrho}$ is suboptimal but consistency and the interior-solution argument are unchanged.

1. Define one routine $\varrho \mapsto (C_t, I_t, Y_t) \mapsto m(\varrho)$ that reuses the trained surrogate.
2. Evaluate the criterion on a fine ϱ -grid over $[0.50, 0.99]$ (step 0.005, 99 points).
3. Take the grid argmin as $\hat{\varrho}$ – the grid is fine enough that no separate optimizer call is needed.
4. **Verify the solution is interior** ($\hat{\varrho}$ strictly inside the bounds), required for asymptotic theory.

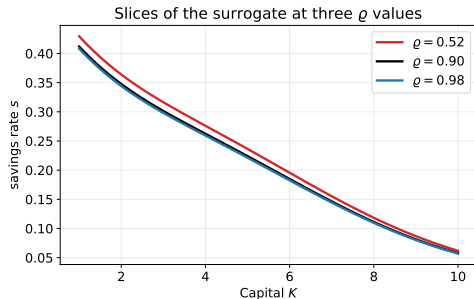
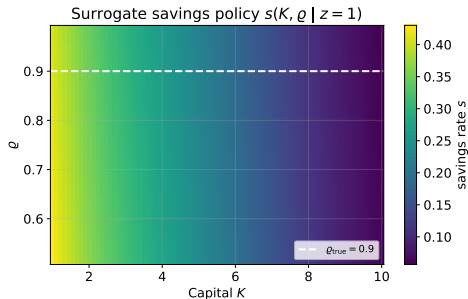
Hint

Re-seed the simulator *to the same seed* on every evaluation, so the candidate parameter is the only thing that changes. Without CRN the criterion is dominated by simulation noise.

Task 3b: Diagnose the Fit (Single-Parameter)

- ▶ How close is $\hat{\varrho}$ to $\varrho_{\text{true}} = 0.90$?
- ▶ Plot the SMM objective profile and mark both ϱ_{true} and $\hat{\varrho}$.
- ▶ Compare matched moments $m(\hat{\varrho})$ versus targets $\hat{m}^{\text{data}}$.
- ▶ Plot $s(K, \varrho_{\text{true}})$ vs. $s(K, \hat{\varrho})$ – the curves should overlay almost exactly.
- ▶ Trace each candidate moment as a function of ϱ on the trained surrogate – this is the identification picture.

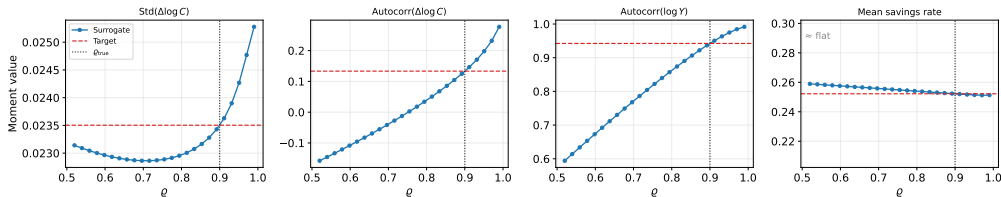
Expected Result: Surrogate Slice $s(K, \varrho \mid z = 1)$



The 3-input network is a smooth function over the entire (K, ϱ) rectangle. The right panel shows three slices; the left panel shows the full surrogate compressed into one heatmap.

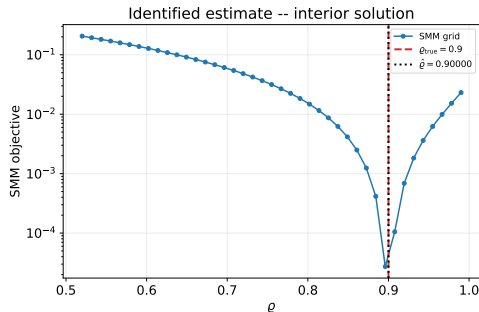
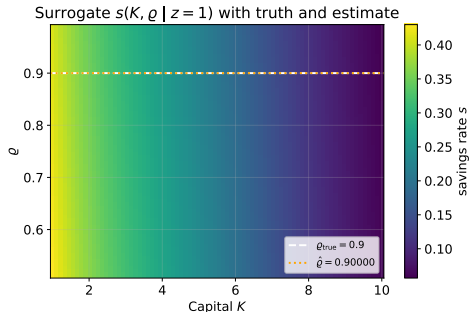
Expected Result: Identifying Variation

Each moment as a function of ϱ at fixed seed (CRN)



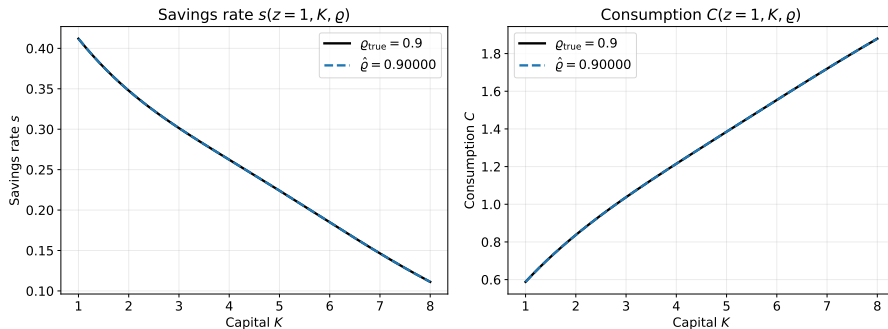
All four candidate moments traced as a function of ϱ : the three identifying moments m_1, m_2, m_3 (Task 2 table, leftmost three panels) are monotone or near-monotone in ϱ . The diagnostic moment $m_0 = \text{mean}(s_t)$ (rightmost panel) is essentially flat – correctly diagnosed as uninformative.

Expected Result: SMM Objective + Estimate



Left: surrogate slice with truth (white dashed) and $\hat{\varrho}$ (orange dotted). Right: log-SMM-objective profile with both. The estimate sits strictly inside the parameter rectangle (**interior** solution).

Expected Result: Policy Match



The estimated policy at $\hat{\rho}$ overlaps the true policy at $\rho_{\text{true}} = 0.90$ to plotting accuracy. Single-parameter SMM with sharp identification recovers the truth accurately under CRN.

Bridge: From Scalar to Joint Estimation

The single-parameter exercise is the cleanest version of the SMM workflow. But two real lessons remain:

- ▶ How does the surrogate scale to *multiple* parameters? Same idea: add them as pseudo-states.
- ▶ What does the SMM criterion look like when one parameter is **partially identified**?

Notebook `02_02_Structural_Estimation_BM_Joint.ipynb` re-trains the surrogate on the rectangle $\beta \in [0.92, 0.99] \times \varrho \in [0.50, 0.99]$ and estimates both jointly.

Two moment subsets:

- ▶ **Weak two-moment** (2 moments): $\{\text{Std}(\Delta \log C), \text{Corr}(\Delta \log C)\}$ – chosen to expose a shallow direction in the criterion.
- ▶ **Over-identified** (4 moments): adds mean savings and $\text{Corr}(\log Y)$.

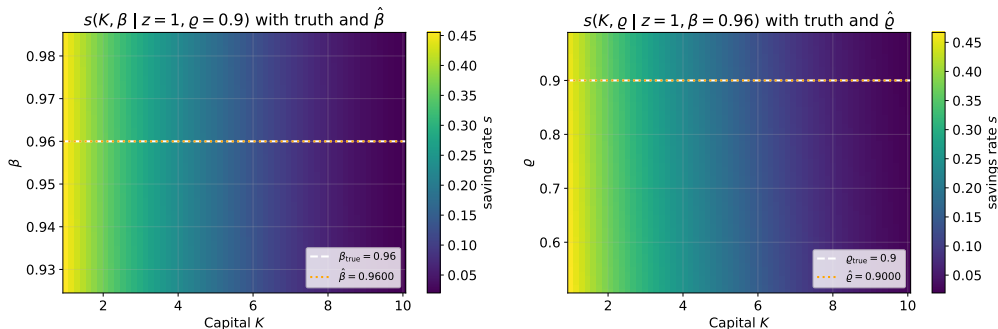
Task 4: Joint Estimation of (β, ϱ)

Goal

$$\hat{\theta} = \arg \min_{\theta=(\beta, \varrho)} [m(\theta) - \hat{m}^{\text{data}}]^\top W [m(\theta) - \hat{m}^{\text{data}}] \quad \text{on } \beta \in [0.92, 0.99] \times \varrho \in [0.50, 0.99].$$

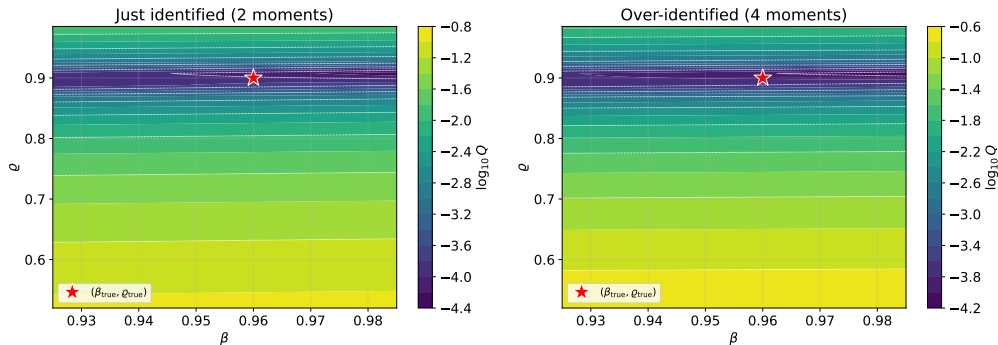
1. 4-input MLP $(z, K, \beta, \varrho) \rightarrow s$. Train once on the whole rectangle.
2. Forge synthetic data at $(\beta_{\text{true}}, \varrho_{\text{true}}) = (0.96, 0.90)$.
3. Compute the SMM criterion on a 29×50 grid in (β, ϱ) – visualize both moment subsets.
4. Take the grid argmin as $\hat{\theta}$ – gradient-free (no FP32 finite-difference issues); the grid is fine enough that no separate optimizer call is needed.
5. Verify the solution is interior; report distance to truth on each axis.

Expected Result: Surrogate Slices on (β, ϱ)



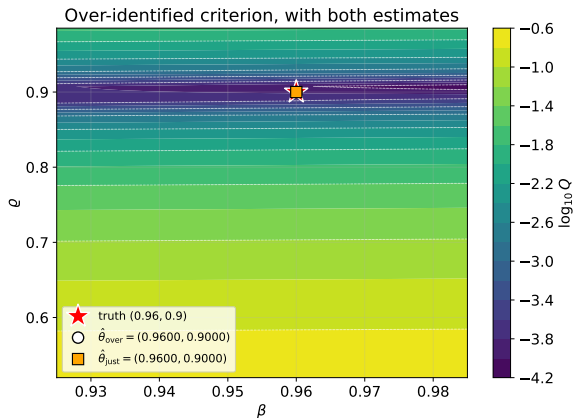
Two slices of the same 4-input network. White dashed = truth, orange dotted = over-identified estimate. The estimate sits strictly **inside** the rectangle on both axes simultaneously.

Expected Result: 2D SMM Criterion



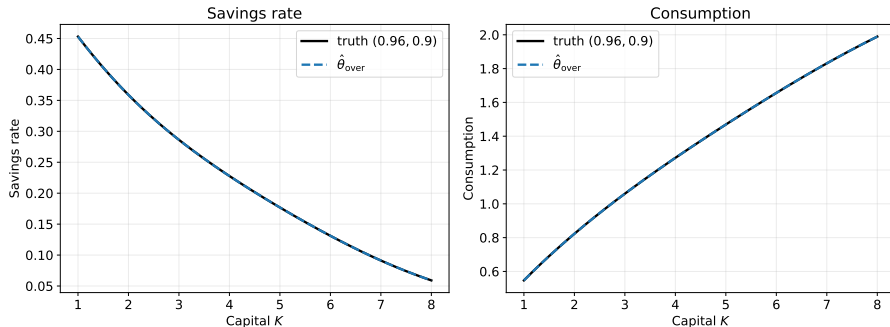
Left: the weak two-moment criterion shows a **ridge** – the partial-identification problem that motivates calibrating β in practice. Right: adding mean savings and output autocorrelation collapses the ridge to a localized minimum.

Expected Result: Estimates on Top of Criterion



In this synthetic CRN run, both estimators recover (β, ρ) close to the truth. The takeaway is the criterion's *shape*: the just-identified ridge would translate into wide confidence bands on β in real-data applications, while the over-identified criterion has localized curvature on both axes.

Joint Policy Match



Savings rate and consumption at the over-identified estimate overlay the truth almost exactly. The 4-input surrogate, trained *once*, supports millions of cheap moment evaluations during the optimizer's line-search.

Wrap-Up

What the two notebooks teach together:

- ▶ A neural surrogate with parameters as pseudo-states **decouples** the cost of solving the structural model from the cost of moment-matching. Train once, use it inside any optimizer.
- ▶ For a well-identified parameter (ϱ), SMM under CRN gives a sharp 1D identification picture.
- ▶ For **joint** estimation, the contour plot makes the partial-identification ridge visible. Adding moments collapses it.
- ▶ The **interior-solution** check is part of the standard pipeline, needed for the asymptotic distribution theory of the SMM estimator.
- ▶ **Synthetic-data caveat.** Exact recovery in both notebooks reflects self-consistency: the data are generated by the same surrogate at the true parameter, and every candidate evaluation reuses the same shock path. Real-data SMM does not produce a zero criterion.

Formal identification machinery (moment Jacobian $M = \partial m / \partial \theta$, rank condition, J -statistic, weak-identification diagnostics): see lecture script Sec. 10.4 (Practical Considerations).

Production pipeline: Chen, Didisheim & Scheidegger (2026, *J. Financial Economics*), exactly this surrogate-and-SMM logic at scale, applied to finance.

Next up (Session 3): the same surrogate machinery answers *design* questions — constrained optimal carbon taxes — and, in the appendix of that deck, *quantify* questions (uncertainty quantification of the social cost of carbon).